Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/05/2007
PI Name:	Collins, Lance R Ph.D.		
Project Title:	Investigation of Particle Clustering Using 3D Digital Holographic Imaging and Direct Numerical Simulations		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	14853	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	09/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received no-cost extension through 9/30/2007, per PI (3/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Meng, Hui (SUNY Buffalo)		
Grant/Contract No.:	NNC05GA45G		
Performance Goal No.:			
Performance Goal Text:			
	It is proposed to develop digital holographic particle image velocimetry (DHPIV) that will make it possible to measure the three-dimensional position of approximately 10^5 particles with resolutions below 10 microns in all three directions. The instrument will be used to investigate particle clustering in turbulent flows. It is well known from direct numerical simulations (DNS) that particles, owing to their higher density, are 'centrifuged' out of regions of high vorticity and collect in regions of high strain. DNS further suggests this clustering can dramatically affect fundamental aerosol processes such as coagulation, vaporization/condensation and dispersion. However, there is limited experimental verification of the DNS observations to date, in large part due to the challenge of measuring particles in turbulence with sufficient accuracy to test the DNS. This proposal addresses this deficiency. In addition to the proposed cutting-edge experimental work, we will perform state-of-the-art DNS that %perfectly complement the experiments. The proposed		

Task Description:	DNS will use a code recently developed in our group that is capable of resolutions of 1024^3 on a 32-node 64-processor Beowulf cluster. We believe simulations with up to 2048^3 can be achieved on the SUNY-Buffalo Center for Computational Research (CCR) cluster, that has more than 300-nodes. The proposed simulations will be the largest multiphase simulations performed to date, and will enable closure of open questions concerning the Reynolds number dependence of particle clustering. The results have important implications for high-Reynolds-number applications such as astrophysics and atmospheric aerosols. In addition, the code has unique capabilities for simulating homogeneous shear flows, enabling us to explore the impact of this parameter on particle clustering, for the first time. The over-arching goal of this proposed study is to develop and validate a unique, next-generation experimental facility and DNS capability to investigate statistical properties of particles in turbulence. DHPIV has the potential to impact a broad range of experimental programs sponsored by NASA that would benefit from 3D particle position and velocity measurements from a single camera. Achieving this ambitious goal requires an interdisciplinary approach. The experimental facility and DHPIV will be developed by Prof. Meng, who has pioneered the use of holographic imaging in turbulence for the last ten years. The DNS will be performed by Prof. Collins, who recently has developed a new parallel algorithm that will enable unprecedented aerosol simulations. Each group brings complementary viewpoints and expertise that will benefit from the other, creating an exciting synergistic team to study turbulence. Any single investigator, quite simply, cannot accomplish the goals of this project.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research involves the integration of direct numerical simulations and high resolution, three-dimensional holographic particle imaging. We are studying particle clustering in turbulence, a phenomenon thought to play an important role in the evolution of clouds on earth, and these in turn affect the weather and global climate change. This project is directed towards understanding the role of atmospheric turbulence on droplet clustering and cloud formation. Additionally, the project is developing more advanced simulation tools for particle-laden turbulence (in particular the extension to homogeneous turbulent shear flow). On the experimental side, we continue to push holographic imaging technology using digital cameras and our own image processing software that we continue to develop.
Task Progress:	The particle code has been coupled to our new numerical Navier Stokes solver (1). In addition gravitational settling of the particles has been incorporated. The code has been generalized to account for a distribution of particle sizes that is matched to the experimenal distribution. In addition to the numerical work we have completed, our group has been working closely with the SUNY-Buffalo group to address the difficult problem of measuring the viscous dissipation rate in the interior of the box. Their group is using two-dimensional PIV to obtain the velocity field over the center of the box; however, the spatial resolution of the technique is insufficient to directly estimate velocity derivatives. Consequently, indirect estimates based on the longitudinal structure function, DLL(r), are used. We then turned our attention to making comparisons of the radial distribution function (RDF) from experiments and DNS. The RDF is a useful measure of particle clustering since its value is the ratio of the rate of collision of clustered particles to the rate of collision in the absence of clustering (2). The particle size distribution in the DNS was matched to the experimental size distribution. From the comparison with the experiment, we see the filters in the range St > 0.05–0.1 work best. We will apply this filter to all subsequent comparisons. The results of this work will be presented as part of a keynote lecture at the ASME Fluids Engineering meeting in Miami in July 2006 (3). References [1] K. A. Brucker, J. C. Isaza, T. Vaithianathan, and L. R. Collins. Efficient algorithm for simulating homogeneous turbulent shear flow without remeshing. J. Comp. Phys., 2006. in review. [2] S. Sundaram and L. R. Collins. Collision statistics in an isotropic, particle-laden turbulent suspension i. direct numerical simulations. J. Fluid Mech., 335:75–109, 1997. [3] Sathyanarayana Ayyalasomayajula, Armann Gylfason, Lujie Cao, Jeremy de Jong, Juan Salazar, Scott Woodward, Lance R. Collins, Zellman Warhaft, Eberhard Bodenschatz, and Hui Meng. Numerical and experimental investigation of particle accelerations and clustering in isotropic turbulence. In ASME FEDSM2006, Miami, 2006. ASME.
Bibliography Type:	Description: (Last Updated: 10/05/2009)
Articles in Peer-reviewed Journals	Brucker KA, Isaza JC, Vaithianathan T, Collins LR. "Efficient algorithm for simulating homogeneous turbulent shear flow without remeshing." Journal of Computational Physics, in press, March 2007. , Mar-2007
Papers from Meeting Proceedings	Brucker KA, Isaza JC, Vaithianathan T, Collins LR. "Numerical and experimental investigation of particle accelerations and clustering in isotropic turbulence." ASME Fluids Engineering Conference, Miami, FL, July 17-20, 2006. ASME Fluids Engineering Conference, July 2006. , Jul-2006